

FinWealth Advisors

Azure Data Warehouse End-to-End Project Report
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Azure Data Factory | Azure SQL Database | Key Vault

Staging → DWH | SCD Type 1 & Type 2 | Star Schema

Version 1.0 | FinWealth DW Team

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01 Project Overview & Architecture

FinWealth Advisors required a centralised Azure Data Warehouse to support wealth analytics dashboards. This document is the complete step-by-step guide covering every resource, script, pipeline, and validation check built during the project.

Architecture Flow

The data pipeline follows a standard Medallion-style two-layer architecture:

Layer	Technology	Description
Source	Azure Blob Storage (raw-data container)	11 CSV files uploaded manually
Staging	Azure SQL — stg schema	Raw landing zone, all VARCHAR, no constraints
DWH	Azure SQL — dwh schema	Star schema with dimensions and facts, typed columns
Orchestration	Azure Data Factory	3 pipelines: Ingest, Transform, Master
Secrets	Azure Key Vault	All passwords and connection strings stored securely

Star Schema Design

The DWH follows a Star Schema with 5 dimension tables and 6 fact tables:

Table	Type	SCD	Source File	Rows
dim_advisor	Dimension	Type 1	Advisors.csv	350
dim_benchmark	Dimension	Type 1	Benchmarks.csv	6 (distinct)
dim_security	Dimension	Type 1	Securities.csv	3,000
dim_client	Dimension	Type 2	Clients.csv	8,000
dim_account	Dimension	Type 2	Accounts.csv	12,000
dim_date	Dimension	Static	Generated	5,479
fact_holding	Fact	—	Holdings.csv	96,027
fact_trade	Fact	—	Trades.csv	100,400
fact_cashflow	Fact	—	Cashflows.csv	30,000
fact_fee	Fact	—	Fees.csv	18,000
fact_security_price	Fact	—	Prices.csv	301,500
fact_performance	Fact	—	Performance.csv	110,000

02 Azure Resource Setup

All resources are created via Azure Portal GUI. Use the same region for all resources to avoid latency and egress costs.

2.1 Resource Group

Navigate to: Azure Portal → Resource groups → + Create

Field	Value
Subscription	Your Azure subscription
Resource group name	rs-finwealth-redo
Region	Central Asia (East India)

2.2 Storage Account & Blob Container

Navigate to: Storage accounts → + Create

Field	Value
Resource group	rs-finwealth-redo
Storage account name	safinwealthredo
Region	West India
Performance	Standard
Redundancy	LRS — Locally Redundant Storage

After creation: Go to Containers → + Container → Name: raw-data → Access: Private. Upload all 11 CSV files into this container.

2.3 Azure Key Vault

Navigate to: Key vaults → + Create. Key Vault stores all secrets so no passwords are hardcoded in ADF.

Field	Value
Resource group	rs-finwealth-redo
Key vault name	kv-finwealth
Region	West India
Pricing tier	Standard

After creation, add the following secrets under Secrets → + Generate/Import:

Secret Name	Value
secretsqlpsw	SQL admin password

2.4 Azure SQL Server & Database

Navigate to: SQL servers → + Create

Field	Value
Server name	sqlsrvrfinwealthredo
Region	Central India
Authentication	SQL Authentication
Admin login	finwealth_admin
Password	Strong password — store in Key Vault

After server creation, create the database: sql-finwealth-dw-prod → + Create database

Field	Value
Database name	sqldbfinwealthredo
Compute + storage	General Purpose, 2 vCores
Backup redundancy	Locally redundant

NOTE: After database creation go to SQL server → Networking → Add your client IP + enable 'Allow Azure services'. Without this ADF cannot connect.

2.5 Azure Data Factory

Navigate to: Data factories → + Create

Field	Value
Resource group	rs-finwealth-redo
Name	adf-finwealth-redo
Region	Central India
Version	V2

2.6 ADF Managed Identity & Key Vault Access Policy

ADF uses its system-assigned Managed Identity to read secrets from Key Vault — no API keys needed.

- Go to adf-finwealth-redo → Properties → copy Managed Identity Object ID
- Go to kv-finwealth → Access policies → + Create
- Secret permissions: tick Get and List only
- Principal: paste the ADF Managed Identity Object ID
- Click Next → Next → Create

2.7 GitHub Integration

Go to ADF Studio → Manage → Git configuration → Configure → GitHub

Field	Value
Repository name	ETL-project
Collaboration branch	main
Publish branch	adf_publish

03 Source Files Analysis

All 11 source CSV files were analysed before any pipeline was built. Key findings:

File	Rows	Columns	Key Issues Found
Advisors.csv	350	5	1 NULL Region → stored as 'Unknown'
Clients.csv	8,000	7	66 NULL RiskProfile → stored as 'Unknown'
Accounts.csv	12,000	6	Clean
Securities.csv	3,000	7	672 NULL Sector → stored as 'N/A' (Bonds/Cash expected)
Benchmarks.csv	4,020	5	Clean — 6 distinct benchmark codes
Cashflows.csv	30,000	5	Signed amounts — positive=inflow, negative=outflow
Fees.csv	18,000	5	All positive amounts — fees always a cost
Holdings.csv	96,027	5	Single snapshot date: 2025-10-31
Performance.csv	110,000	7	109 NULL ReturnPct → loaded as NULL
Prices.csv	301,500	3	478 NULL ClosePrice → loaded as NULL
Trades.csv	100,400	9	250 negative settle lags; 60 SELL with positive amount (source anomaly)

04 SQL Database Setup

All scripts are run in Azure Portal → SQL databases → sqldbfinwealthredo → Query editor. Run each statement separately.

4.1 Schema Creation

Run each line separately in Query Editor:

```
CREATE SCHEMA stg;  
CREATE SCHEMA dwh;
```

4.2 Staging Tables

Staging tables use all VARCHAR columns — this prevents load failures due to type mismatches. A stg_load_timestamp column is added for audit purposes.

12 staging tables created:

Staging Table	Source File	Key Columns
stg.stg_advisors	Advisors.csv	AdvisorID, AdvisorName, Region, JoinDate, ExperienceYears
stg.stg_clients	Clients.csv	ClientID, ClientName, Segment, RiskProfile, PrimaryAdvisorID
stg.stg_accounts	Accounts.csv	AccountID, ClientID, AccountType, OpenDate, BaseCurrency, Status
stg.stg_securities	Securities.csv	SecurityID, Ticker, SecurityName, AssetClass, Sector, Currency
stg.stg_benchmarks	Benchmarks.csv	BenchmarkCode, Description, Currency, IndexValueDate, IndexValue
stg.stg_cashflows	Cashflows.csv	CashflowID, AccountID, CashflowDate, Type, Amount
stg.stg_fees	Fees.csv	FeeID, AccountID, FeeDate, FeeType, Amount
stg.stg_holdings	Holdings.csv	AccountID, SecurityID, Quantity, ValuationDate, MarketValue
stg.stg_performance	Performance.csv	AccountID, MonthStart, ReturnPct, BenchmarkCode
stg.stg_prices	Prices.csv	SecurityID, PriceDate, ClosePrice
stg.stg_trades	Trades.csv	TradeID, AccountID, SecurityID, TradeDate, Side, Price, Quantity

4.1 DWH Dimension Tables — SCD Type 1 & 2

SCD Type 1 (Overwrite): dim_advisor, dim_benchmark, dim_security — changes overwrite existing record, history not needed.

SCD Type 2 (Full History): dim_client, dim_account — changes create a new row, old row is expired with effective_end_date and is_current = N.

Dimension	SCD Type	Surrogate Key	Natural Key	Special Handling
dim_advisor	Type 1	advisor_sk	advisor_id	NULL Region → 'Unknown'
dim_benchmark	Type 1	benchmark_sk	benchmark_code	Clean
dim_security	Type 1	security_sk	security_id	NULL Sector → 'N/A'
dim_client	Type 2	client_sk	client_id	NULL RiskProfile → 'Unknown'; effective_start/end dates
dim_account	Type 2	account_sk	account_id	Status/type changes tracked; effective_start/end dates
dim_date	Static	date_sk	full_date	YYYYMMDD integer key; covers 2012-01-01 to 2026-12-31

4.2 DWH Fact Tables

Fact tables use surrogate keys from dimension tables. Natural keys are retained in fact tables for traceability.

Fact Table	Grain	Key FKs	Notes
fact_holding	Account + Security + Date	account_sk, security_sk, date_sk	AUM source; full refresh each run
fact_trade	One row per trade	account_sk, security_sk, trade_date_sk, settle_date_sk	settle_lag_days derived column added
fact_cashflow	One row per cashflow	account_sk, date_sk	is_inflow Y/N derived from amount sign
fact_fee	One row per fee	account_sk, date_sk	All amounts positive
fact_security_price	Security + Date	security_sk, date_sk	478 NULL prices allowed
fact_performance	Account + Benchmark + Month	account_sk, benchmark_sk, date_sk	109 NULL return_pct allowed

4.3 Audit Log Table

All pipeline runs are logged to dbo.audit_pipeline_log:

```
CREATE TABLE dbo.audit_pipeline_log (  
  log_id INT IDENTITY(1,1) PRIMARY KEY,  
  pipeline_name VARCHAR(100) NOT NULL,  
  source_file VARCHAR(100) NOT NULL,  
  target_table VARCHAR(100) NOT NULL,
```

```
run_start_time DATETIME NOT NULL,  
run_end_time DATETIME NULL,
```

```
rows_loaded INT NULL,  
status VARCHAR(20) NOT NULL, -- RUNNING / SUCCESS / FAILED  
error_message VARCHAR(2000) NULL,  
loaded_by VARCHAR(100) DEFAULT 'ADF'  
);
```

05 ADF Configuration

5.1 Linked Services

Three Linked Services connect ADF to external resources. All secrets fetched from Key Vault — no hardcoded passwords.

Linked Service Name	Type	Connection	Secret Used
Is_keyvault_finwealth	Azure Key Vault	kv-finwealth-dw-prod	—
LSAzureBlobStorageFinwealthredo	Azure Blob Storage	safinwealthredo	secret-blob-finwealth-accesskey
LSAzureSqlDatabaseFinwealthredo	Azure SQL Database	sqldbfinwealthredo	secret-sql-finwealth-password

NOTE: Always create Is_keyvault_finwealth first. The other two linked services reference it for their secret values.

5.2 Datasets — CSV Source (ds_csv folder)

11 DelimitedText datasets pointing to Azure Blob Storage. First row as header = checked. Schema imported from connection.

Dataset Name	File	Linked Service
ds_csv_advisors	Advisors.csv	LSAzureBlobStorageFinwealthredo
ds_csv_clients	Clients.csv	LSAzureBlobStorageFinwealthredo
ds_csv_accounts	Accounts.csv	LSAzureBlobStorageFinwealthredo
ds_csv_securities	Securities.csv	LSAzureBlobStorageFinwealthredo
ds_csv_benchmarks	Benchmarks.csv	LSAzureBlobStorageFinwealthredo
ds_csv_cashflows	Cashflows.csv	LSAzureBlobStorageFinwealthredo
ds_csv_fees	Fees.csv	LSAzureBlobStorageFinwealthredo
ds_csv_holdings	Holdings.csv	LSAzureBlobStorageFinwealthredo
ds_csv_performance	Performance.csv	LSAzureBlobStorageFinwealthredo
ds_csv_prices	Prices.csv	LSAzureBlobStorageFinwealthredo
ds_csv_trades	Trades.csv	LSAzureBlobStorageFinwealthredo

5.3 Datasets — SQL Sink (ds_sql folder)

11 Azure SQL Database datasets pointing to staging tables.

Dataset Name	Table	Linked Service
ds_sql_stg_advisors	stg.stg_advisors	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_clients	stg.stg_clients	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_accounts	stg.stg_accounts	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_securities	stg.stg_securities	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_benchmarks	stg.stg_benchmarks	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_cashflows	stg.stg_cashflows	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_fees	stg.stg_fees	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_holdings	stg.stg_holdings	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_performance	stg.stg_performance	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_prices	stg.stg_prices	LSAzureSqlDatabaseFinwealthredo
ds_sql_stg_trades	stg.stg_trades	LSAzureSqlDatabaseFinwealthredo

NOTE: When mapping columns in Copy activities, delete the stg_load_timestamp row from the mapping. It is auto-filled by SQL DEFAULT GETDATE() and has no CSV source column.

06 ADF Pipelines

Three pipelines are created across two folders. The master pipeline calls both pipelines in sequence with one click.

Pipeline	Purpose
pl_ingest_finwealth_allsources	Truncate all staging tables, copy all 11 CSVs in parallel, log result
pl_transform_stg_to_dwh	Load all dimensions then facts via stored procedures, log result
Master_pipeline_finwealth	Execute ingest then transform end-to-end with single Debug/Trigger

6.1 pl_ingest_all_csv_to_stg

Pipeline flow:

- Activity 1 — script_truncate_all_stg (Script activity): Truncates all 11 staging tables in one SQL block
- Activities 2-12 — 11 Copy Data activities all connected from truncate script with green arrows, running in PARALLEL
- Activity 13 — sp_log_success (Stored Procedure): Connected from all 11 copy activities with GREEN arrow
- Activity 14 — sp_log_failure (Stored Procedure): Connected from all 11 copy activities with RED arrow

NOTE: Delete the stg_load_timestamp mapping row in each Copy activity's Mapping tab before publishing.

6.2 pl_transform_stg_to_dwh

Pipeline flow — strict sequential order for dimensions, then parallel for facts:

- sp_load_dim_advisor → sp_load_dim_benchmark → sp_load_dim_security → sp_load_dim_client → sp_load_dim_account (sequential green arrows)
- sp_load_dim_account → all 6 fact SPs in PARALLEL (6 green arrows out)
- All 6 fact SPs → sp_log_transform_success (GREEN arrows)
- All 6 fact SPs → sp_log_transform_failure (RED arrows)

SP Activity Name	Stored Procedure	Type
sp_load_dim_advisor	dwh.usp_load_dim_advisor	SCD Type 1
sp_load_dim_benchmark	dwh.usp_load_dim_benchmark	SCD Type 1
sp_load_dim_security	dwh.usp_load_dim_security	SCD Type 1
sp_load_dim_client	dwh.usp_load_dim_client	SCD Type 2
sp_load_dim_account	dwh.usp_load_dim_account	SCD Type 2
sp_load_fact_holding	dwh.usp_load_fact_holding	Full Refresh

SP Activity Name	Stored Procedure	Type
sp_load_fact_trade	dwh.usp_load_fact_trade	Full Refresh
sp_load_fact_cashflow	dwh.usp_load_fact_cashflow	Full Refresh
sp_load_fact_fee	dwh.usp_load_fact_fee	Full Refresh
sp_load_fact_security_price	dwh.usp_load_fact_security_price	Full Refresh
sp_load_fact_performance	dwh.usp_load_fact_performance	Full Refresh

6.3 pl_master_finwealth_dwh

Single pipeline with 2 Execute Pipeline activities:

- exec_pl_ingest_all_csv_to_stg — Invoked pipeline: pl_ingest_all_csv_to_stg — Wait on completion: ON
- exec_pl_transform_stg_to_dwh — Invoked pipeline: pl_transform_stg_to_dwh — Wait on completion: ON
- Connect with ONE green arrow: exec_pl_ingest → exec_pl_transform

NOTE: Running pl_master_finwealth_dwh once completes the full end-to-end load from CSV files to DWH star schema.

07 Stored Procedures

7.1 Dimension SPs — SCD Logic Explained

SCD Type 1 (dim_advisor, dim_benchmark, dim_security): UPDATE existing record if any tracked column has changed. INSERT new records that do not yet exist. No history preserved — latest values overwrite.

SCD Type 2 (dim_client, dim_account): If a tracked column changes, the existing row is EXPIRED (effective_end_date = today, is_current = N). A new row is INSERTED with effective_start_date = today, effective_end_date = NULL, is_current = Y. Both old and new versions are retained.

SP Name	Schema	SCD	Key Logic
usp_load_dim_advisor	dwh	Type 1	UPDATE changed; INSERT new; NULL Region → 'Unknown'
usp_load_dim_benchmark	dwh	Type 1	UPDATE changed; INSERT new (DISTINCT on BenchmarkCode)
usp_load_dim_security	dwh	Type 1	UPDATE changed; INSERT new; NULL Sector → 'N/A'
usp_load_dim_client	dwh	Type 2	Expire changed rows; insert new version; insert brand new
usp_load_dim_account	dwh	Type 2	Expire changed rows; insert new version; insert brand new

7.2 Fact SPs — Full Refresh Strategy

All fact SPs use TRUNCATE + full reload strategy. This is appropriate because source files contain complete history each time. Surrogate key lookups use is_current = Y for SCD-2 dimensions.

SP Name	TRUNCATE	Key Joins	Special Logic
usp_load_fact_holding	Yes	dim_account (is_current=Y), dim_security, dim_date	MarketValue = AUM source
usp_load_fact_trade	Yes	dim_account, dim_security, dim_date x2 (trade+settle)	settle_lag_days = DATEDIFF(trade,settle)
usp_load_fact_cashflow	Yes	dim_account, dim_date	is_inflow derived from amount sign
usp_load_fact_fee	Yes	dim_account, dim_date	All amounts positive
usp_load_fact_security_price	Yes	dim_security, dim_date	NULL ClosePrice loaded as NULL

SP Name	TRUNCATE	Key Joins	Special Logic
usp_load_fact_performance	Yes	dim_account, dim_benchmark, dim_date	NULL ReturnPct loaded as NULL

7.3 Audit Logging SPs

SP Name	Schema	Called By	Parameters
usp_log_ingest_summary	dbo	pl_ingest_all_csv_to_stg (success+failure SP)	status, error_message
usp_log_transform_summary	dbo	pl_transform_stg_to_dwh (success+failure SP)	status, error_message

08 Data Quality & Validation

Validation Checks Performed

Check	Result	Status
Audit log entries	2 SUCCESS entries (ingest + transform)	✓ Pass
Staging row counts match source CSV	All 11 tables exact match	✓ Pass
DWH row counts match staging	All 11 tables exact match	✓ Pass
dim_advisor NULL region	1 record → stored as 'Unknown'	✓ Pass
dim_client NULL risk_profile	66 records → stored as 'Unknown'	✓ Pass
dim_security NULL sector	672 records → stored as 'N/A' (expected)	✓ Pass
SCD Type 2 — dim_client current rows	8,000 current records	✓ Pass
SCD Type 2 — dim_account current rows	12,000 current records	✓ Pass
Referential integrity — all facts	Zero orphan records across all 6 facts	✓ Pass
Cashflow sign rules	Zero violations — all types correctly signed	✓ Pass
NULL ClosePrice in fact_security_price	478 NULLs loaded as NULL (documented)	✓ Pass
NULL ReturnPct in fact_performance	109 NULLs loaded as NULL (documented)	✓ Pass
Negative settle lag trades	250 source anomalies loaded as-is	✓ Pass
SELL trades with positive amount	60 source anomalies loaded as-is	✓ Pass

Data Quality Rules Applied

Rule	Column	Source Value	DWH Value
NULL Region	dim_advisor.region	NULL	'Unknown'
NULL RiskProfile	dim_client.risk_profile	NULL or empty	'Unknown'
NULL Sector	dim_security.sector	NULL or empty	'N/A'
NULL ClosePrice	fact_security_price.close_price	NULL	NULL (allowed, documented)
NULL ReturnPct	fact_performance.return_pct	NULL	NULL (allowed, documented)

Validation SQL Queries

Run these queries anytime to verify data health:

```
-- Row count verification
SELECT 'dim_advisor', COUNT(*) FROM dwh.dim_advisor UNION ALL
SELECT 'dim_client', COUNT(*) FROM dwh.dim_client UNION ALL
SELECT 'fact_holding',COUNT(*) FROM dwh.fact_holding;

-- Referential integrity check
SELECT 'orphan holdings', COUNT(*) FROM dwh.fact_holding WHERE account_sk IS NULL
UNION ALL
SELECT 'orphan trades', COUNT(*) FROM dwh.fact_trade WHERE account_sk IS NULL;
```

09 Naming Conventions

All objects follow consistent naming conventions throughout the project:

Object Type	Convention	Example
Resource Group	rs	rsfinwealthredo
Storage Account	sa	safinwealthredo
Key Vault	kv	kv-finwealth
SQL Server	sqlsrvr	sqlsrvrfinwealthredo
SQL Database	sqldb	sqldbfinwealthredo
Data Factory	adf	adf-finwealthredo
ADF Linked Service	ls	LSAzureBlobStorageFinwealthredo
ADF Dataset CSV	ds_csv	ds_csv_advisors
ADF Dataset SQL	ds_sql_stg	ds_sql_stg_advisors
ADF Pipeline	pl	pl_ingest_all_csv_to_stg
Key Vault Secret	secret	secret-sql-finwealth-password
Staging Table	stg.	stg.stg_advisors
Dimension Table	dwh.	dwh.dim_advisor
Fact Table	dwh.	dwh.fact_holding
Stored Procedure	dwh.	dwh.usp_load_dim_advisor

10 Expected Row Counts

Use these counts to verify a successful end-to-end pipeline run:

Table	Layer	Expected Rows	Status
stg.stg_advisors	Staging	350	✓ Pass
stg.stg_clients	Staging	8,000	✓ Pass
stg.stg_accounts	Staging	12,000	✓ Pass
stg.stg_securities	Staging	3,000	✓ Pass
stg.stg_benchmarks	Staging	4,020	✓ Pass
stg.stg_cashflows	Staging	30,000	✓ Pass
stg.stg_fees	Staging	18,000	✓ Pass
stg.stg_holdings	Staging	96,027	✓ Pass
stg.stg_performance	Staging	110,000	✓ Pass
stg.stg_prices	Staging	301,500	✓ Pass
stg.stg_trades	Staging	100,400	✓ Pass
dwh.dim_advisor	DWH	350	✓ Pass
dwh.dim_benchmark	DWH	6	✓ Pass
dwh.dim_security	DWH	3,000	✓ Pass
dwh.dim_client	DWH	8,000	✓ Pass
dwh.dim_account	DWH	12,000	✓ Pass
dwh.fact_holding	DWH	96,027	✓ Pass
dwh.fact_trade	DWH	100,400	✓ Pass
dwh.fact_cashflow	DWH	30,000	✓ Pass
dwh.fact_fee	DWH	18,000	✓ Pass
dwh.fact_security_price	DWH	301,500	✓ Pass
dwh.fact_performance	DWH	110,000	✓ Pass

Total rows in DWH: 683,297 across 11 DWH tables (6 dimensions + 6 facts).

11 Screenshots

11.1 Resources

☐	Name ↑	Type	Location
☐	adf-finwealth-redo	... Data factory (V2)	Central India
☐	kv-finwealth	... Key vault	West India
☐	safinwealthredo	... Storage account	West India
☐	sqldbfinwealthredo (sqlsrvfinwealthredo/sqldbfinwealthredo)	... SQL database	Central India
☐	sqlsrvfinwealthredo	... SQL server	Central India

11.2 PL-TRANSFORM STG TO DWH

Microsoft Azure | Data Factory | ADF-finwealth | Validate all resources and publish them | Search | kachhiainish@gmail.com | DEFAULT DIRECTORY

main branch | Validate all | Save all | Publish | Migrate to Fabric (Preview)

Factory Resources

- Pipelines (3)
 - PL_ingest_Finnwealth_AllSources
 - PL_Master_Pipeline_FinnWealth
 - PL_Transform_stg_to_DWH
- Change Data Capture (preview) (0)
- Datasets (22)
 - Ds_csv (11)
 - Ds_FinnWealth_Accounts
 - Ds_FinnWealth_Advisors
 - Ds_FinnWealth_Benchmarks
 - Ds_FinnWealth_Cashflows
 - Ds_FinnWealth_Clients
 - Ds_FinnWealth_Fees
 - Ds_FinnWealth_Holdings
 - Ds_FinnWealth_Performance
 - Ds_FinnWealth_Prices
 - Ds_FinnWealth_Securities
 - Ds_FinnWealth_Trades

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Parameters | Variables | Settings | Output

+ New

11.3 INGEST_FINWEALTH_ALLSOURCES

Microsoft Azure | Data Factory | ADF-finwealth | Validate all resources and publish them | Search | kachhiainish@gmail.com | DEFAULT DIRECTORY

main branch | Validate all | Save all | Publish | Migrate to Fabric (Preview)

Factory Resources

- Pipelines (3)
 - PL_ingest_Finnwealth_AllSources
 - PL_Master_Pipeline_FinnWealth
 - PL_Transform_stg_to_DWH
- Change Data Capture (preview) (0)
- Datasets (22)
 - Ds_csv (11)
 - Ds_FinnWealth_Accounts
 - Ds_FinnWealth_Advisors
 - Ds_FinnWealth_Benchmarks
 - Ds_FinnWealth_Cashflows
 - Ds_FinnWealth_Clients
 - Ds_FinnWealth_Fees
 - Ds_FinnWealth_Holdings
 - Ds_FinnWealth_Performance
 - Ds_FinnWealth_Prices
 - Ds_FinnWealth_Securities
 - Ds_FinnWealth_Trades

Activities

- Move and transform
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Parameters | Variables | Settings | Output

+ New

11.4 MASTER PIPELINE

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar includes the Microsoft Azure logo, the breadcrumb path 'Data Factory > ADF-finnwealth', and a search bar. The left sidebar shows the 'Factory Resources' tree with categories like Pipelines, Datasets, and DS_csv. The main workspace is divided into three panes: 'Activities' on the left, a central canvas showing a pipeline flow with 'Execute Pipeline' tasks, and a 'Parameters' pane at the bottom. The pipeline flow consists of two 'Execute Pipeline' tasks: 'PL_Ingest_FinnWealth' followed by 'PL_Transform_stg_to_DWH'.

11.5 BLOB STORAGE CONTAINER

The screenshot shows the Microsoft Azure portal interface for a Blob Storage container named 'raw-data'. The left sidebar contains the 'All services' list. The main area displays the container's overview, including a search bar, a list of blobs, and a table of blob details. The table lists 11 items, each with a checkbox, name, last modified date, access tier, blob type, size, and lease state.

	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Accounts.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	596.35 KiB	Available
☐	Advisors.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	14.63 KiB	Available
☐	Benchmarks.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	180.32 KiB	Available
☐	Cashflows.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	1.35 MiB	Available
☐	Clients.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	552.48 KiB	Available
☐	Fees.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	837.88 KiB	Available
☐	Holdings.csv	5/22/2026, 1:18:33 PM	Hot (Inferred)	Block blob	4.22 MiB	Available
☐	Performance.csv	5/22/2026, 1:18:33 PM	Hot (Inferred)	Block blob	6.18 MiB	Available
☐	Prices.csv	5/22/2026, 3:25:03 PM	Hot (Inferred)	Block blob	9.94 MiB	Available
☐	Securities.csv	5/22/2026, 1:18:32 PM	Hot (Inferred)	Block blob	168.18 KiB	Available

11.6 DWH TABLE CONTENTS EXAMPLE (QUERY EDITOR)

Home > Resource Manager | Resource groups > rs-finwealth-redo > sqlsruvfinwealthredo | SQL databases > sqldbfinwealthredo (sqlsruvfinwealthredo/sqldbfinwealthredo)

sqldbfinwealthredo (sqlsruvfinwealthredo/sqldbfinwealthredo) | Query editor (preview) ☆ ...

Search

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dwh

Tables

dim_account

dim_advisor

dim_benchmark

fact_cashflow

dim_account

Data preview - dim_account Showing 1000 rows Search

	123	account_sk	ABC account_id	ABC client_id	ABC account...	open_date	ABC base_curr...	ABC status
1	1		AC0000001	CL003967	Joint	2016-06-26	USD	Active
2	2		AC0000002	CL000978	Joint	2016-11-19	GBP	Active
3	3		AC0000003	CL002947	Individual	2015-02-22	EUR	Active
4	4		AC0000004	CL007091	Individual	2015-10-16	EUR	Active
5	5		AC0000005	CL001583	Retirement	2022-06-10	USD	Closed
6	6		AC0000006	CL001638	Individual	2014-12-21	GBP	Active
7	7		AC0000007	CL007978	Individual	2013-03-11	USD	Active
8	8		AC0000008	CL002767	Joint	2017-03-05	USD	Active
9	9		AC0000009	CL005550	Individual	2017-07-16	EUR	Active
10	10		AC0000010	CL001283	Individual	2012-06-05	JPY	Active
11	11		AC0000011	CL005035	Joint	2024-04-17	USD	Active
12	12		AC0000012	CL004588	Individual	2015-02-19	GBP	Active
13	13		AC0000013	CL000102	Corporate	2019-11-24	USD	Closed
14	14		AC0000014	CL006333	Retirement	2021-11-13	USD	Active
15	15		AC0000015	CL005298	Individual	2025-02-04	USD	Active

Succeeded (0 sec 200 ms) Columns: 12 Rows: 1000

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11.7.1 VALIDATION

Home > Resource Manager | Resource groups > rs-finwealth-redo > sqlsruvfinwealthredo | SQL databases > sqldbfinwealthredo (sqlsruvfinwealthredo/sqldbfinwealthredo)

sqldbfinwealthredo (sqlsruvfinwealthredo/sqldbfinwealthredo) | Query editor (preview) ☆ ...

Search

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dim_account

dim_advisor

dim_benchmark

fact_cashflow

dim_account

SQL query 1

Run Save as view

```
1 SELECT 'dim_advisor' , COUNT(*) FROM dwh.dim_advisor UNION ALL
2 SELECT 'dim_benchmark' , COUNT(*) FROM dwh.dim_benchmark UNION ALL
3 SELECT 'dim_security' , COUNT(*) FROM dwh.dim_security UNION ALL
4 SELECT 'dim_client' , COUNT(*) FROM dwh.dim_client UNION ALL
5 SELECT 'dim_account' , COUNT(*) FROM dwh.dim_account UNION ALL
```

Messages Results

	ABC	123
1	dim_advisor	350
2	dim_benchmark	6
3	dim_security	3000
4	dim_client	8000
5	dim_account	12000
6	fact_holding	96027
7	fact_trade	100000
8	fact_cashflow	30000
9	fact_fee	18000
10	fact_security_price	301500

Succeeded (2 sec 360 ms) Columns: 2 Rows: 11

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11.7.2 (REFER FEATURED QUERY FOR BETTER UNDERSTANDING)

Home › Resource Manager | Resource groups › rs-finwealth-redo › sqlsrvfinwealthredo | SQL databases › sqldbfinwealthredo (sqlsrvfinwealthredo/sqldbfinwealthredo)

SQL database

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fact_cashflow dim_account SQL query 1

Run Save as view

```
1 SELECT 'dim_advisor NULL region' , COUNT(*) FROM dwh.dim_advisor WHERE region = 'Unknown'
2 UNION ALL
3 SELECT 'dim_client NULL risk' , COUNT(*) FROM dwh.dim_client WHERE risk_profile = 'Unknown'
4 UNION ALL
5 SELECT 'dim_security NULL sector' , COUNT(*) FROM dwh.dim_security WHERE sector = 'N/A'
6 UNION ALL
7 -- Check SCD Type 2 current records
8 SELECT 'dim_client current records' , COUNT(*) FROM dwh.dim_client WHERE is_current = 'Y'
9 UNION ALL
10 SELECT 'dim_account current records' , COUNT(*) FROM dwh.dim_account WHERE is_current = 'Y';
```

Messages Results

	ABC	123
1	dim_advisor NULL region	1
2	dim_client NULL risk	66
3	dim_security NULL sector	672
4	dim_client current records	8000
5	dim_account current records	12000

Succeeded (0 sec 118 ms) Columns: 2 Rows: 5

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11.7.3

fact_cashflow dim_account SQL query 1

Run Save as view

```
8 UNION ALL
9 SELECT 'fact_cashflow orphan accounts' , COUNT(*) FROM dwh.fact_cashflow WHERE account_sk IS NULL
10 UNION ALL
11 SELECT 'fact_fee orphan accounts' , COUNT(*) FROM dwh.fact_fee WHERE account_sk IS NULL
12 UNION ALL
13 SELECT 'fact_price orphan securities' , COUNT(*) FROM dwh.fact_security_price WHERE security_sk IS NULL
14 UNION ALL
15 SELECT 'fact_perf orphan accounts' , COUNT(*) FROM dwh.fact_performance WHERE account_sk IS NULL
16 UNION ALL
17 SELECT 'fact_perf orphan benchmarks' , COUNT(*) FROM dwh.fact_performance WHERE benchmark_sk IS NULL;
```

Messages Results

	ABC	123
1	fact_holding orphan accounts	0
2	fact_holding orphan securities	0
3	fact_trade orphan accounts	0
4	fact_trade orphan securities	0
5	fact_cashflow orphan accounts	0
6	fact_fee orphan accounts	0

Succeeded (0 sec 134 ms) Columns: 2 Rows: 9

11.7.4 AUM BY CURRENCY

fact_cashflow

dim_account

SQL query 1

Run

Save as view

```
44 -- Total AUM by currency
45 SELECT
46     da.base_currency,
47     COUNT(DISTINCT fh.account_id) AS total_accounts,
48     SUM(fh.market_value) AS total_aum,
49     AVG(fh.market_value) AS avg_holding_value
50 FROM dwh.fact_holding fh
51 INNER JOIN dwh.dim_account da
52     ON da.account_sk = fh.account_sk
```

Messages

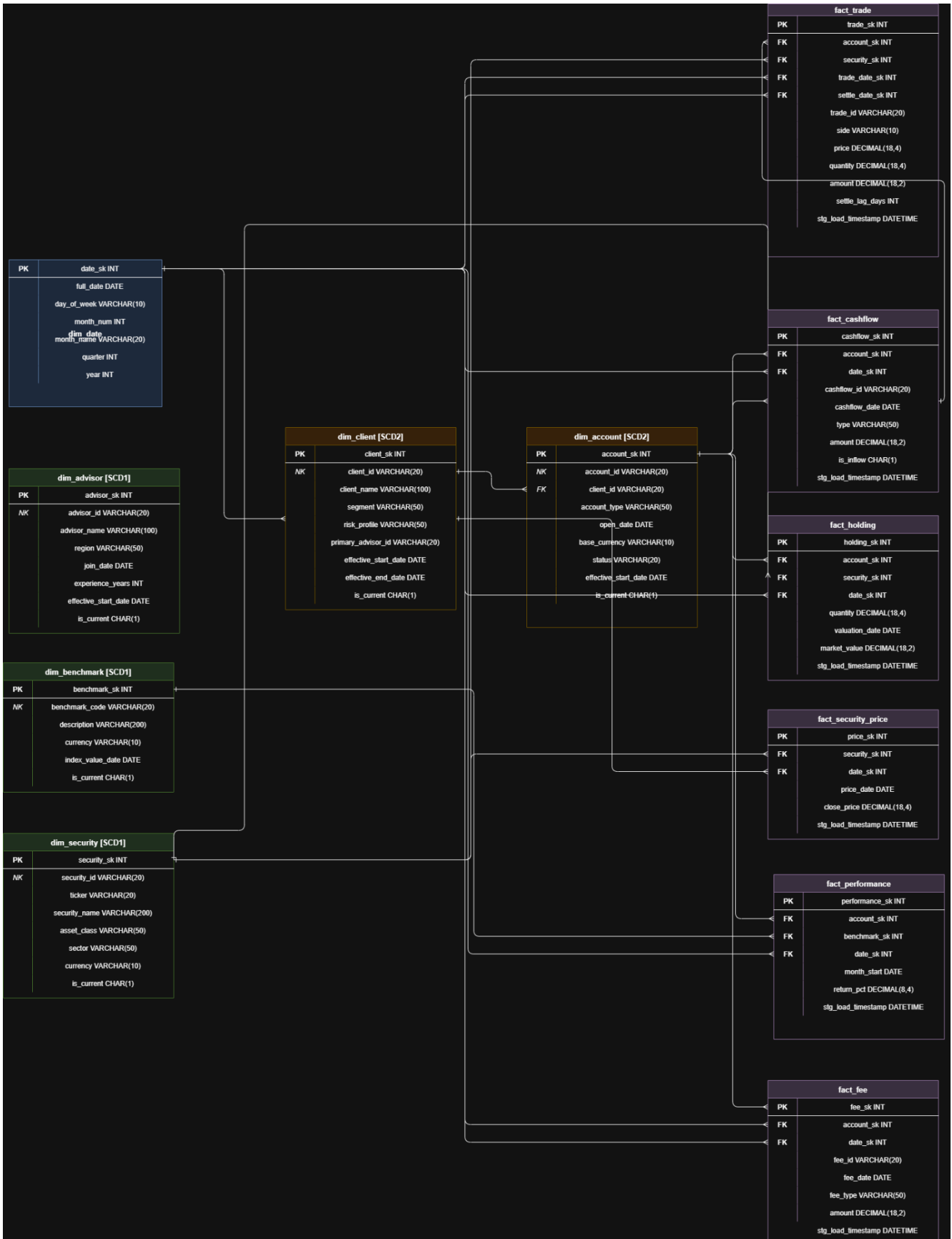
Results

	ABC base_currency	123 total_accounts	e* total_aum	e* avg_holding_value
1	USD	7254	1793842682.34	30876.152059
2	GBP	1197	286470159.28	29956.097383
3	EUR	1149	276637004.06	30200.546294
4	JPY	633	152820379.58	30273.450788
5	CAD	583	145128821.59	30767.187108
6	AUD	569	142152150.50	31030.812158
7	SGD	349	84055306.17	30377.775992

✓ Succeeded (3 sec 448 ms)

Columns: 4 Rows: 9

12 ER Diagram



Legend

1. **Blue** — dim_date (static)
2. **Green** — dim_advisor, dim_benchmark, dim_security (SCD Type 1)
3. **Orange** — dim_client, dim_account (SCD Type 2)
4. **Purple** — all 6 fact tables